

SABIR HUSNAIN

Mechatronics & Control Engineer



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<https://envisionpath.net>sabirhasnain577@gmail.com[linkedin.com/in/sabir-husnain-awan](https://www.linkedin.com/in/sabir-husnain-awan)github.com/SabirHusnain[Sabir Husnain – Upwork](#)**SUMMARY**

Passionate and driven Mechatronics and Control Engineer with a strong academic background. My expertise spans across diverse engineering fields, including mechanical, electronics, control systems, embedded systems, robotics, computer vision, and machine learning. My true fascination lies in the domains of control, robotics, computer vision, and machine learning where I aspire to become a dedicated research scientist contributing to cutting-edge technology advancements. My determination, adaptability, and problem-solving abilities enable me to tackle complex challenges with ease.

RESEARCH INTEREST

- Robotics: Design, Control, Manipulation, Perception and Training
- Autonomous Vehicles, Unmanned Aerial Robots
- Robust and Adaptive Control Systems
- Non Linear Control Design for MIMO Systems
- Deep Learning: Intelligent Control, Reinforcement Learning, Manipulation

EDUCATION

Oct, 2019 – Sep, 2023

BE Mechatronics & Control Engineering**University of Engineering & Technology Lahore, Pakistan**

▪ CGPA = 3.793 / 4.0

▪ **Thesis:** Development of low cost portable optical motion capture system for assessment of gross motor movements – BIGKAT (Boxed Infrared Gross Kinematic Assessment Tool)

▪ **Achievements:** Full fee concession: Merit based scholarship from HEC, Degree with honours, Student of Session 2019-23.

WORK EXPERIENCE

Feb, 2024 – Present

Design Engineer Team Lead**Landlele SRL, Italy****Full-Time**

- Development of hybrid solar inverters, charge controllers and green energy devices.
- Managing a team of engineers working on the development of devices.
- Development and Implementation innovative engineering ideas in the domain.
- Conducting R&D for the exploration of solutions to problems being faced by company in engineering domain.

Nov, 2023 – Present

Chief Mechatronics Engineer**SD Companies SRL, Altamura, Italy****Full-Time**

- Proposing and executing project plans smoothly and efficiently.
- Development of technical plans for industrial project.
- Managing industrial projects.
- Managing and leading team of engineers in projects development.
- Leading R&D for higher efficiency and better project designs.

May, 2023 – Nov, 2023

Process Optimization Engineer**SD Companies SRL, Altamura, Italy****Full-Time**

- Streamline cutting machines for improved efficiency and productivity.
- Develop autonomous cutting, grinding, and welding robotic systems.
- Test and deploy software for seamless robotic system integration.
- Conduct in-depth analysis of manufacturing processes to identify areas for improvement.
- Implement advanced control strategies to optimize cutting, grinding, and welding operations.
- Perform risk assessments and develop safety protocols for robotic systems.
- Continuously monitor and fine-tune robotic systems to ensure peak performance.
- Participate in R&D projects to explore new process optimization possibilities.

Oct, 2022 – Feb, 2023

Automation, Control Engineer**Hydroequip LLC, Dubai, UAE****Contract**

- Programmed PLCs for hydraulic & pneumatics-based systems control.
- Implemented robotics simulations for validation.
- Developed, tested, and maintained precise robotic manipulators.
- Devised innovative control algorithms and networking interfaces.
- Collaborated to design integrated automation solutions.
- Conducted feasibility studies for automation projects.
- Implemented real-time monitoring and predictive maintenance.
- Ensured safety protocols for robotic operations.

May, 2022 – Sep, 2022

Advanced Embedded System Engineer**EPTeck, Faisalabad, Pakistan****Part-Time**

- Developed Embedded Systems using Arduino, ESP32, STM32, PIC, ARM, Raspberry Pi, and FPGA.
- Used ESP-IDF, FreeRTOS, and ZephyrRTOS frameworks for efficient system development.
- Designed and optimized Embedded System circuits, PCBs, and FPGA configurations for peak performance.
- Created computer applications tailored to enhance embedded system functionalities.
- Developed mobile applications to seamlessly integrate with embedded systems.
- Orchestrated plant automation and control solutions.

May, 2021 – Current

Mechatronics Engineer & Founder**Self Employed****EnvisionPath**

- Machine Vision, Artificial Intelligence and Machine Learning.
- Robotics (Mechanical Design, Electronics, Control Design).
- PLC & DCS Automation with IIOT.
- Control System Development.
- Embedded Linux Image Developer (Yocto Project).
- Embedded System Development (FPGA, STM32, ESP-IDF, RTOS, Arduino, Esp32, Teensy, PIC).
- Modelling & Simulation of Mechatronics System using CAD and MATLAB etc.
- Mechanical Design and Analysis (CAD, ANSYS, FEA).
- CNC Machine Programming and CAD / CAM 3D Modelling in Fusion 360, Inventor, AutoCAD and SolidWorks.
- Application development using .NET, C, C++, C#, JAVA, Python and JavaScript.
- Electric & Electronic Circuit Design, PCB Design and Simulations.

PUBLICATIONS**Accepted**

- Husnain, S., Hafeez, A., Systematic Review: 2D and 3D Kinematic Assessment of Upper Limb Motor Dysfunction in PD Patients and its Correlation with Clinical Outcome Measures.

In Review

- S. Saeed, S. Husnain, A Comparison of H_∞ (H-infinity) and ADRC (Active Disturbance Rejection Control) for a MIMO System with Uncertainty.
- S. Saeed, S., Husnain, S., Emebu. Development of power optimized Energy Saving Fan with Sensorless FOC Control.

Published

- S. Husnain and R. Abdulkader, "Fractional Order Modeling and Control of an Articulated Robotic Arm", Eng. Technol. Appl. Sci. Res., vol. 13, no. 6, pp. 12026–12032, Dec. 2023.
- Zarghoon, S., Emebu, S., Matušů, R., Belavý, C., Bartalský, L., Ďuriš, S., Husnain, S. and Martinez, C.M., 2024. Full-state feedback LQR with integral gain for control of induction heating of steel billet. Engineering Science and Technology, an International Journal, 55, p.101721.
- S. Husnain, M. T. Riaz, M. A. Khan and M. Mazhar, "An Artificial Bee Colony Based PID Tuning Algorithm for Inverted Pendulum and Cart System," 2023 17th International Conference on Open Source Systems and Technologies (ICOSST), Lahore, Pakistan, 2023, pp. 1-7, doi: 10.1109/ICOSST60641.2023.10414233.
- Riaz, M.T., Hashmi, W.S., Ahmad, S., Husnain, S., Mujtaba, H., Ali, H., Atiq, S. and Qureshi, M.M., 2022, March. Design of Optimization Methodology for Economic Dispatch of Thermal Generating Units. In *2022 5th International Conference on Energy Conservation and Efficiency (ICECE)* (pp. 1-5). IEEE. doi: 10.1109/ICECE54634.2022.9758966.
- Riaz, M.T., Rehman, W.U., Ali, H., Husnain, S., Jiang, G., Lodhi, E., Aaqib, S.M. and Qureshi, M.M., 2021, October. Design and Experimental Validation of a Small-Scale Prototype Active Aerostatic Thrust Bearing. In *2021 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube)* (pp. 1-6). IEEE. doi: 10.1109/ICECube53880.2021.9628270.
- Riaz, M.T., Aaqib, S.M., Ahmad, S., Amin, S., Ali, H., Husnain, S. and Riaz, S., 2021, October. The Intelligent Transportation Systems with Advanced Technology of Sensor and Network. In *2021 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube)* (pp. 1-6). IEEE. doi: 10.1109/ICECube53880.2021.9628331.

SKILLS**Engineering & Software Skills****Robotics/Computer Vision/Machine Learning**

ROS/ROS2 (C++/Python) | Control System Design | Motion Planning | Autonomous Navigation | Robot Manipulation & Grasping | CoppeliaSim | WeBots | MATLAB | Expert Systems | Deep Learning (Neural Networks, Sequence Models, Auto Encoders) | Reinforcement Learning | Large Language Models

Mechatronics

Mechatronics Design | Modelling & Simulation | Mechanical Design | CAD/CAM | Ansys | Digital Signal Processing | Product Design

Control System Design

Dynamic System Modelling | Classical Control (PID, State Feedback, LQR, LQI, Feed-Forward, SMO) | Non Linear Control | Robust Control (H_∞ , μ -Synthesis, Loop Shaping) | Adaptive and Data Driven Control (MPC, ADRC, MRAC, Gain Scheduling) | Intelligent Control (Fuzzy/Neural Network/Evolutionary) | Control Hardware Design

Embedded Systems & Electronics

ARM | Atmel | STM | PIC | Teensy | RTOS | ESP-IDF | DSP | FPGA | Embedded Linux | Raspberry Pi | PCB Design (Altium, KiCAD, EagleCAD)

Industrial Automation

PLC (Siemens, Fatek, Delta, Allen Bradley, Omron) | SCADA | DCS | IIOT

Programming, Databases & IT

C | C++ | C# | Python | .NET | Qt | SQLite | MySQL | Oracle | Git | Linux

Power Skills

- Public speaking, Excellent communication and Presentation skills
- Team lead
- Proficient, Imaginative visualization
- Problem solving

PROJECTS**Academic****BIGKAT (Boxed Infrared Gross Kinematic Assessment Tool)**

- Low-cost, portable motion capture system for assessment of gross motor movements.
- Uses active infrared markers to track the motion and provide analysis such as 3D coordinate, Maximum Grip Aperture (MGA), distance between markers, motion time stamps.
- Machine learning model analyse for Bradykinesia by comparing the results with control models.
- Python (PyQt5) based GUI including touch screen for user-friendly interface.
- Project won competition for Ignite – National Technology Fund's funding.

Link: <https://github.com/sabirhusnain577/BIGKAT.git> (Private – Access on Request – FYP)

Autonomous mobile robot using artificial neural networks and genetic algorithm

- Differential drive robot simulated in MATLAB environment with dynamics in consideration.
- Implemented a feed-forward neural network and PID control to process ultrasonic sensors' outputs.
- Employed genetic algorithm to optimize the neural network's parameters for optimized path.

Link: <https://github.com/sabirhusnain577/NnGa-Robot.git>

PID, LQR & state feedback control of inverted pendulum and cart using low-cost IMU

- Dynamic modelling and parameter estimation of inverted pendulum and cart system considering the motors and environmental factors such as friction, damping, inertia.
- Design and tuning of controllers in simulation environment such as MATLAB.
- Development of firmware for IMU (Accelerometer and Gyro) with gravity compensation.
- Development of filters and control system on hardware (esp32 microcontroller).

Link: <https://github.com/sabirhusnain577/IP-Control.git>

Humanoid robot control and path generation using reinforcement learning

- Simulation of a humanoid robot control system and path generation using reinforcement learning.
- Employing reinforcement-learning techniques to train the humanoid robot's control policy.
- Generating optimal paths for the robot's movement through the environment using the learned control policy.
- The reinforcement learning approach enables the robot to adapt and improve its control strategy in real-time based on feedback from the environment.

Link: <https://github.com/sabirhusnain577/RL-Human.git>

Humanoid robot (Sponsored by Ahmad Jamal Textile, Faisalabad Pvt. Ltd.)

- Design and development of humanoid fashion robot for a textile industry to test and display their fabric quality.
- Fabrication of humanoid robot and fitting metallic links in a mannequin.
- Controlled using model based non-linear feed forward controller along with PID.
- Walking trajectory planning for robot by employing robot dynamics and point of zero moment.

4-DOF robotics manipulator with optimized trajectory generation

- Employed MATLAB to simulate the system using sim-mechanics and robotics system toolbox.
- Designed dynamic controller for robot along with PID controllers.
- Obstacle avoidance path planning using Probabilistic Road Maps (PRM) and A* Algorithm.
- Cubic polynomial interpolation for trajectory generation.

Link: <https://github.com/sabirhusnain577/Torque-Optimized-Arm.git>

5-DOF trainable robotic manipulator

- Design & development of small-scale servo system based robotic manipulator that can be trained.
- Development of mathematical model of robot on hardware, and writing firmware to store position feedback for training process.
- Modified commercial servos to get a position feedback signal.
- Robotic control firmware along with inter-system communication.

Link: <https://github.com/sabirhusnain577/5-DOF-Trainable-Robotic-Manipulator.git>

Python application for Artificial Reverberation

- A digital signal processing-based python application using PyQt5 GUI library to simulate the effect of artificial reverberation.

Link: <https://github.com/sabirhusnain577/Artificial-Reverberations.git>

Professional**Design & simulation of wheat bale handling mobile manipulator**

- Designed 6-DOF robotic manipulator, mobile robot and custom gripper for wheat bale handling.
- Design of low-level control, trajectory control, and supervisory logic for mobile robot, 6-DOF robotic manipulator and gripper.
- Development of simulation environment on MATLAB with physical interaction between environment and robot (real time physics).
- Path and trajectory planning for both systems using PRM and cubic polynomial interpolation trajectory

planning.

Link: <https://github.com/sabirhusnain577/MM-Robot.git> (Private – Access on Request)

PID controlled underwater robot using Artificial Bee Colony (ABC)

- Mathematical modelling of kinematics and non-linear dynamics of underwater mobile robot.
- Designing 2-DOF PID controller for non-linear model obtained.
- Development of simulation environment and ABC algorithm on MATLAB for the system
- Controller tuning using ABC algorithm and finding optimized parameters.

Link: <https://github.com/sabirhusnain577/ABC-UAV.git>

Other Projects

- Hydraulic circuit design, control and PLC automation for garbage truck, HydraoEquip, Dubai.
- RemoteWell framework and hardware circuit design, EPTeck.
- BMS and VCU design for electric vehicle for a German automobile manufacturer.

CERTIFICATIONS & ACHIEVEMENTS

- Top 2nd Mechatronics Engineer providing services on Upwork in the world.
- Deep Learning Specialization (Stanford University Online, 5 Certifications)
- Machine Learning Specialization (Stanford University Online, 3 Certifications)
- Robotics Specialization (University of Pennsylvania, 5 Certifications)
- Modern Robotics Specialization (Northwestern University, 6 Certifications)
- Python IT Automation Specialization (Google, 7 Certifications)
- Ignite project funding for development of BIGKAT
- HEC project funding for development and research on BIGKAT
- Presenting and research author for The Design of Optimization Methodology for Economic Dispatch of Thermal Generating Units (Research Author - IEEE)
- Presenting and research author for The Design and Experimental Validation of a Small-Scale Prototype Active Aerostatic Thrust Bearing (Research Author - IEEE)
- Presenting and research author for The Transportation Systems with Advanced Technology of Sensor and Network (Research Author - IEEE)

REFERENCES

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